APPENDIX D

The Constellations

Latin Name	Genitive	Abbrev.	Translation	R. A. h	Dec. deg
Andromeda	Andromedae	And	Princess of Ethiopia	1	+40
Antlia	Antliae	Ant	Air Pump	10	−35
Apus	Apodis	Aps	Bird of Paradise	16	−75
Aquarius	Aquarii	Aqr	Water Bearer	23	−15
Aquila	Aquilae	Aql	Eagle	20	+5
Ara	Arae	Ara	Altar	17	−55
Aries	Arietis	Ari	Ram	3	+20
Auriga	Aurigae	Aur	Charioteer	6	+40
Boötes	Boötis	Boo	Herdsman	15	+30
Caelum	Caeli	Cae	Chisel	5	−40
Camelopardalis	Camelopardis	Cam	Giraffe	6	+70
Cancer	Cancri	Cnc	Crab	9	+20
Canes Venatici	Canum Venaticorum	CVn	Hunting Dogs	13	+40
Canis Major	Canis Majoris	CMa	Big Dog	7	−20
Canis Minor	Canis Minoris	CMi	Little Dog	8	+5
Capricornus	Capricorni	Cap	Goat	21	−20
Carina	Carinae	Car	Ship’s Keel	9	−60
Cassiopeia	Cassiopeiae	Cas	Queen of Ethiopia	1	+60
Centaurus	Centauri	Cen	Centaur	13	−50
Cepheus	Cephei	Cep	King of Ethiopia	22	+70
Cetus	Ceti	Cet	Sea Monster (whale)	2	−10
Chamaeleon	Chamaeleontis	Cha	Chameleon	11	−80
Circinus	Circini	Cir	Compass	15	−60
Columba	Columbae	Col	Dove	6	−35
Coma Berenices	Comae Berenices	Com	Berenice’s Hair	13	+20
Corona Australis	Coronae Australis	CrA	Southern Crown	19	−40
Corona Borealis	Coronae Borealis	CrB	Northern Crown	16	+30
Corvus	Corvi	Crv	Crow	12	−20
Crater	Crateris	Crt	Cup	11	−15
Crux	Crucis	Cru	Southern Cross	12	−60
Cygnus	Cygni	Cyg	Swan	21	+40
Delphinus	Delphini	Del	Dolphin, Porpoise	21	+10
Dorado	Doradus	Dor	Swordfish	5	−65
Draco	Draconis	Dra	Dragon	17	+65
Equuleus	Equulei	Equ	Little Horse	21	+10
Eridanus	Eridani	Eri	River Eridanus	3	−20
Fornax	Fornacis	For	Furnace	3	−30
Gemini	Geminorum	Gem	Twins	7	+20
Grus	Gruis	Gru	Crane	22	−45
Hercules	Herculis	Her	Son of Zeus	17	+30

Latin Name	Genitive	Abbrev.	Translation	R. A. h	Dec. deg
Horologium	Horologii	Hor	Clock	3	−60
Hydra	Hydrae	Hya	Water Snake	10	−20
Hydrus	Hydri	Hyi	Sea Serpent	2	−75
Indus	Indi	Ind	Indian	21	−55
Lacerta	Lacertae	Lac	Lizard	22	+45
Leo	Leonis	Leo	Lion	11	+15
Leo Minor	Leonis Minoris	LMi	Little Lion	10	+35
Lepus	Leporis	Lep	Hare	6	−20
Libra	Librae	Lib	Balance, Scales	15	−15
Lupus	Lupi	Lup	Wolf	15	−45
Lynx	Lyncis	Lyn	Lynx	8	+45
Lyra	Lyrae	Lyr	Lyre, Harp	19	+40
Mensa	Mensae	Men	Table, Mountain	5	−80
Microscopium	Microscopii	Mic	Microscope	21	−35
Monoceros	Monocerotis	Mon	Unicorn	7	−5
Musca	Muscae	Mus	Fly	12	−70
Norma	Normae	Nor	Square, Level	16	−50
Octans	Octantis	Oct	Octant	22	−85
Ophiuchus	Ophiuchi	Oph	Serpent-bearer	17	0
Orion	Orionis	Ori	Hunter	5	+5
Pavo	Pavonis	Pav	Peacock	20	−65
Pegasus	Pegasi	Peg	Winged Horse	22	+20
Perseus	Persei	Per	Rescuer of Andromeda	3	+45
Phoenix	Phoenicis	Phe	Phoenix	1	−50
Pictor	Pictoris	Pic	Painter, Easel	6	−55
Pisces	Piscium	Psc	Fish	1	+15
Piscis Austrinus	Piscis Austrini	PsA	Southern Fish	22	−30
Puppis	Puppis	Pup	Ship's Stern	8	−40
Pyxis	Pyxidis	Pyx	Ship's Compass	9	−30
Reticulum	Reticuli	Ret	Net	4	−60
Sagitta	Sagittae	Sge	Arrow	20	+10
Sagittarius	Sagittarii	Sgr	Archer	19	−25
Scorpius	Scorpii	Sco	Scorpion	17	−40
Sculptor	Sculptoris	Scl	Sculptor	0	−30
Scutum	Scuti	Sct	Shield	19	−10
Serpens	Serpentis	Ser	Serpent	17	0
Sextans	Sextantis	Sex	Sextant	10	0
Taurus	Tauri	Tau	Bull	4	+15
Telescopium	Telescopii	Tel	Telescope	19	−50
Triangulum	Trianguli	Tri	Triangle	2	+30
Triangulum Australe	Trianguli Australis	TrA	Southern Triangle	16	−65
Tucana	Tucanae	Tuc	Toucan	0	−65
Ursa Major	Ursae Majoris	UMa	Big Bear	11	+50
Ursa Minor	Ursae Minoris	UMi	Little Bear	15	+70
Vela	Velorum	Vel	Ship's Sai	9	−50
Virgo	Virginis	Vir	Maiden, Virgin	13	0
Volans	Volantis	Vol	Flying Fish	8	−70
Vulpecula	Vulpeculae	Vul	Little Fox	20	+25

APPENDIX

E The Brightest Stars

Name	Star	Spectral Class A	Spectral Class B	V^a A	V^a B	M_V A	M_V B
Sirius	α CMa	A1 V	wd[b]	-1.44	$+8.7$	$+1.45$	$+11.6$
Canopus	α Car	F0 Ib		-0.62		-5.53	
Arcturus	α Boo	K2 II Ip		-0.05		-0.31	
Rigel Kentaurus	α Cen	G2 V	K0 V	-0.01	$+1.3$	$+4.34$	$+5.7$
Vega	α Lyr	A0 V		$+0.03$		$+0.58$	
Capella[c]	α Aur	G8 III	G0 III	$+0.08$	$+10.2$	-0.48	$+9.5$
Rigel	β Ori	B8 Ia	B9	$+0.18$	$+6.6$	-6.69	-0.4
Procyon	α CMi	F5 IV–V	wd[b]	$+0.40$	$+10.7$	$+2.68$	$+13.0$
Betelgeuse	α Ori	M2Ib		$+0.45$v		-5.14	
Achernar	α Eri	B3 Vp		$+0.45$		-2.77	
Hadar	β Cen	B1 III	?	$+0.61$	$+4$	-5.42	-0.8
Altair	α Aql	A7 IV–V		$+0.76$		$+2.20$	
Acrux	α Cru	B0.5 IV	B3	$+0.77$	$+1.9$	-4.19	-3.5
Aldebaran	α Tau	K5 III	M2V	$+0.87$	$+13$	-0.63	$+12$
Spica	α Vir	B1 V		$+0.98$v		-3.55	
Antares	α Sco	M1 Ib	B4eV	$+1.06$v	$+5.1$	-5.58	-0.3
Pollux	β Gem	K0 III		$+1.16$		$+1.09$	
Fomalhaut	α PsA	A3 V	K4V	$+1.17$	$+6.5$	$+1.74$	$+7.3$
Deneb	α Cyg	A2 Ia		$+1.25$		-8.73	
Mimosa	β Cru	B0.5 III	B2 V	$+1.25$v		-3.92	

[a] Values labeled v designate variable stars.

[b] wd represents a white dwarf star.

[c] Capella has a third member of spectral class M5 V, $V = +13.7$, and $M_V = +13$.

Name	R. A.[a] (h m s)	Dec.[a] (° ′ ″)	Parallax[b] (″)	Distance[c] (pc)	Proper Motion[d] (″ yr^{-1})	Radial Velocity (km s^{-1})
Sirius	06 45 08.92	−16 41 58.0	0.37921(158)	2.64	1.33942	−7.7
Canopus	06 23 57.11	−52 41 44.4	0.01043(53)	95.88	0.03098	+20.5
Arcturus	14 15 39.67	+19 10 56.7	0.08885(74)	11.26	2.27887	−5.2
Rigel Kentaurus	14 39 36.50	−60 50 02.3	0.74212(140)	1.35	3.70962	−24.6
Vega	18 36 56.34	+38 47 01.3	0.12893(55)	7.76	0.35077	−13.9
Capella	05 16 41.36	+45 59 52.8	0.07729(89)	12.94	0.43375	+30.2
Rigel	05 14 32.27	−08 12 05.9	0.00422(81)	237	0.00195	+20.7
Procyon	07 39 18.12	+05 13 30.0	0.28593(88)	3.50	1.25850	−3.2
Betelgeuse	05 55 10.31	+07 24 25.4	0.00763(164)	131	0.02941	+21.0
Achernar	01 37 42.85	−57 14 12.3	0.02268(57)	44.09	0.09672	+19
Hadar	14 03 49.40	−60 22 22.9	0.00621(56)	161	0.04221	−12
Altair	19 50 47.0	+08 52 06.0	0.19444(94)	5.14	0.66092	−26.3
Acrux	12 26 35.90	−63 05 56.7	0.01017(67)	98.33	0.03831	−11.2
Aldebaran	04 35 55.24	+16 30 33.5	0.05009(95)	19.96	0.19950	+54.1
Spica	13 25 11.58	−11 09 40.8	0.01244(86)	80.39	0.05304	+1.0
Antares	16 29 24.46	−26 25 55.2	0.00540(168)	185	0.02534	−3.2
Pollux	07 45 18.95	+28 01 34.3	0.09674(87)	10.34	0.62737	+3.3
Fomalhaut	22 57 39.05	−29 37 20.1	0.13008(92)	7.69	0.36790	+6.5
Deneb	20 41 25.91	+45 16 49.2	0.00101(57)	990	0.00220	−4.6
Mimosa	12 47 43.26	−59 41 19.5	0.00925(61)	108	0.04991	+10.3

[a] Right ascension and declination are given in epoch J2000.0.
[b] Parallax data are from the Hipparcos Space Astrometry Mission. Uncertainties are in parentheses; for instance, the parallax of Sirius is $0.37921'' \pm 0.00158''$.
[c] Distance was calculated from the parallax measurement.
[d] Proper motion data are from the Hipparcos Space Astrometry Mission.

APPENDIX

F The Nearest Stars

Name	HIP[a]	Spectral Class	V^b	M_V	$B-V$	Parallax[c] ($''$)	Distance[d] (pc)
Proxima Centauri (α Cen C)	70890	M5 Ve	11.01	15.45	+1.81	0.77233(242)	1.29
α Cen B	71681	K1 V	1.35	5.70	+0.88	0.74212(140)	1.35
α Cen A	71683	G2 V	−0.01	4.34	+0.71	0.74212(140)	1.35
Barnard's Star	87937	M5 V	9.54	13.24	+1.57	0.54901(158)	1.82
Gl 411	54035	M2 Ve	7.49	10.46	+1.50	0.39240(91)	2.55
Sirius A (α CMa)	32349	A1 V	−1.44	1.45	+0.01	0.37921(158)	2.64
Sirius B (α CMa)		wd (DA)	8.44	11.33	−0.03	0.37921(158)[e]	2.64[e]
Gl 729	92403	M4.5 Ve	10.37	13.00	+1.51	0.33648(182)	2.97
ϵ Eri	16537	K2 V	3.72	6.18	+0.88	0.31075(85)	3.22
Gl 887	114046	M2 Ve	7.35	9.76	+1.48	0.30390(87)	3.29
Ross 128 (Gl 447)	57548	M4.5 V	11.12	13.50	+1.75	0.29958(220)	3.34
6l Cyg A (Gl 820)	104214	K5 Ve	5.20	7.49	+1.07	0.28713(151)	3.48
Procyon A (α CMi)	37279	F5 IV–V	0.40	2.68	+0.43	0.28593(88)	3.50
Procyon B (α CMi)		wd	10.7	13.0	+0.00	0.28593(88)[e]	3.50[e]
61 Cyg B (Gl 820B)	104217	K7 Ve	6.05	8.33	+1.31	0.28542(72)	3.50
Gl 725B	91772	M5 V	9.70	11.97	+1.56	0.28448(501)	3.52
Gl 725A	91768	M4 V	8.94	11.18	+1.50	0.28028(257)	3.57
GX And	1475	M2 V	8.09	10.33	+1.56	0.28027(105)	3.57
ϵ Ind	108870	K5 Ve	4.69	6.89	+1.06	0.27576(69)	3.63
τ Cet	8102	G8 Vp	3.49	5.68	+0.73	0.27417(80)	3.65
Gl 54.1	5643	M5.5 Ve	12.10	14.25	+1.85	0.26905(757)	3.72
Luyten's Star (Gl 237)	36208	M3.5	9.84	11.94	+1.57	0.26326(143)	3.80
Kapteyn's Star	24186	M0 V	8.86	10.89	+1.55	0.25526(86)	3.92
AX Mic	105090	M0 Ve	6.69	8.71	+1.40	0.25337(113)	3.95
Kruger 60	110893	M2 V	9.59	11.58	+1.61	0.24952(303)	4.01
Ross 614 (GL 234A)	30920	M4.5 Ve	11.12	13.05	+1.69	0.24289(264)	4.12

[a] HIP designates the Hipparcos catalog number.
[b] Values labeled v designate variable stars.
[c] Parallax data are from the Hipparcos Space Astrometry Mission. Uncertainties are in parentheses; for instance, the parallax of Proxima Centauri is $0.77233'' \pm 0.00242''$.
[d] Distances were calculated from the Hipparcos parallax data.
[e] Parallax and distance taken to be that of bright companion.

Name	R. A.[a] (h m s)	dec.[a] (° ′ ″)	Proper Motion R. A.[b] (″ yr^{-1})	Proper Motion dec.[b] (″ yr^{-1})	Radial. Velocity. (km s^{-1})
Proxima Centauri	14 29 42.95	−62 40 46.1	−3.77564(152)	0.76816(182)	−33.4
α Cen B	14 39 35.08	−60 50 13.8	−3.60035(2610)	0.95211(1975)	−23.4
α Cen A	14 39 36.50	−60 50 02.3	−3.67819(151)	0.48184(124)	−23.4
Barnard's Star	17 57 48.50	+04 41 36.2	−0.79784(161)	10.32693(129)	−112.3
Gl 411	11 03 20.19	+35 58 11.6	−0.58020(77)	−4.76709(77)	−85.
Sirius A	06 45 08.92	−16 41 58.0	−0.54601(133)	−1.22308(124)	−7.7
Gl 729	18 49 49.36	−23 50 10.4	0.63755(222)	−0.19247(145)	−7.0
ϵ Eri	03 32 55.84	−09 27 29.7	−0.97644(98)	0.01797(91)	+13.
Gl 887	23 05 52.04	−35 51 11.1	6.76726(70)	1.32666(74)	−6.4
Ross 128	11 47 44.40	+00 48 16.4	0.60562(214)	−1.21923(186)	−31.
61 Cyg A	21 06 53.94	+38 44 57.9	4.15510(95)	3.25890(119)	−65.
Procyon A	07 39 18.12	+05 13 30.0	−0.71657(88)	−1.03458(38)	−3.2
61 Cyg B	21 06 55.26	+38 44 31.4	4.10740(43)	3.14372(59)	−65.
Gl 725B	18 42 46.90	+59 37 36.6	−1.39320(1150)	1.84573(1202)	+1.
Gl 725A	18 42 46.69	+59 37 49.4	−1.32688(310)	1.80212(358)	−1.
GX And	00 18 22.89	+44 01 22.6	2.88892(75)	0.41058(63)	+13.5
ϵ Ind	22 03 21.66	−56 47 09.5	3.95997(55)	−2.53884(42)	−40.4
τ Cet	01 44 04.08	−15 56 14.9	−1.72182(83)	0.85407(80)	−16.4
Gl 54.1	01 12 30.64	−16 59 56.3	1.21009(521)	0.64695(391)	+37.0
Luyten's Star	07 27 24.50	+05 13 32.8	0.57127(141)	−3.69425(90)	+18.
Kapteyn's Star	05 11 40.58	−45 01 06.3	6.50605(95)	−5.73139(90)	+242.8
AX Mic	21 17 15.27	−38 52 02.5	−3.25900(128)	−1.14699(56)	+23.
Kruger 60	22 27 59.47	+57 41 45.1	−0.87023(300)	−0.47110(297)	−34.
Ross 614	06 29 23.40	−02 48 50.3	0.69473(300)	−0.61862(248)	+23.2

[a] Right ascension and declination are given in epoch J2000.0.

[b] Proper-motion data are from the Hipparcos Space Astrometry Mission. Uncertainties are in parentheses; for instance, the proper motion of Proxima Centauri in right ascension is $-3.77564'' \text{ yr}^{-1} \pm 0.00152'' \text{ yr}^{-1}$.

APPENDIX

G Stellar Data

Main-Sequence Stars (Luminosity Class V)

Sp. Type	T_e (K)	$L/L_\odot$	$R/R_\odot$	$M/M_\odot$	$M_{\rm bol}$	BC	M_V	$U-B$	$B-V$
O5	42000	499000	13.4	60	−9.51	−4.40	−5.1	−1.19	−0.33
O6	39500	324000	12.2	37	−9.04	−3.93	−5.1	−1.17	−0.33
O7	37500	216000	11.0	—	−8.60	−3.68	−4.9	−1.15	−0.32
O8	35800	147000	10.0	23	−8.18	−3.54	−4.6	−1.14	−0.32
B0	30000	32500	6.7	17.5	−6.54	−3.16	−3.4	−1.08	−0.30
B1	25400	9950	5.2	—	−5.26	−2.70	−2.6	−0.95	−0.26
B2	20900	2920	4.1	—	−3.92	−2.35	−1.6	−0.84	−0.24
B3	18800	1580	3.8	7.6	−3.26	−1.94	−1.3	−0.71	−0.20
B5	15200	480	3.2	5.9	−1.96	−1.46	−0.5	−0.58	−0.17
B6	13700	272	2.9	—	−1.35	−1.21	−0.1	−0.50	−0.15
B7	12500	160	2.7	—	−0.77	−1.02	+0.3	−0.43	−0.13
B8	11400	96.7	2.5	3.8	−0.22	−0.80	+0.6	−0.34	−0.11
B9	10500	60.7	2.3	—	+0.28	−0.51	+0.8	−0.20	−0.07
A0	9800	39.4	2.2	2.9	+0.75	−0.30	+1.1	−0.02	−0.02
A1	9400	30.3	2.1	—	+1.04	−0.23	+1.3	+0.02	+0.01
A2	9020	23.6	2.0	—	+1.31	−0.20	+1.5	+0.05	+0.05
A5	8190	12.3	1.8	2.0	+2.02	−0.15	+2.2	+0.10	+0.15
A8	7600	7.13	1.5	—	+2.61	−0.10	+2.7	+0.09	+0.25
F0	7300	5.21	1.4	1.6	+2.95	−0.09	+3.0	+0.03	+0.30
F2	7050	3.89	1.3	—	+3.27	−0.11	+3.4	+0.00	+0.35
F5	6650	2.56	1.2	1.4	+3.72	−0.14	+3.9	−0.02	+0.44
F8	6250	1.68	1.1	—	+4.18	−0.16	+4.3	+0.02	+0.52

Main-Sequence Stars (Luminosity Class V)

Sp. Type	T_e (K)	$L/L_\odot$	$R/R_\odot$	$M/M_\odot$	$M_{\rm bol}$	BC	M_V	$U-B$	$B-V$
G0	5940	1.25	1.06	1.05	+4.50	−0.18	+4.7	+0.06	+0.58
G2	5790	1.07	1.03	—	+4.66	−0.20	+4.9	+0.12	+0.63
Sun[a]	5777	1.00	1.00	1.00	+4.74	−0.08	+4.82	+0.195	+0.650
G8	5310	0.656	0.96	—	+5.20	−0.40	+5.6	+0.30	+0.74
K0	5150	0.552	0.93	0.79	+5.39	−0.31	+5.7	+0.45	+0.81
K1	4990	0.461	0.91	—	+5.58	−0.37	+6.0	+0.54	+0.86
K3	4690	0.318	0.86	—	+5.98	−0.50	+6.5	+0.80	+0.96
K4	4540	0.263	0.83	—	+6.19	−0.55	+6.7	—	+1.05
K5	4410	0.216	0.80	0.67	+6.40	−0.72	+7.1	+0.98	+1.15
K7	4150	0.145	0.74	—	+6.84	−1.01	+7.8	+1.21	+1.33
M0	3840	0.077	0.63	0.51	+7.52	−1.38	+8.9	+1.22	+1.40
M1	3660	0.050	0.56	—	+7.99	−1.62	+9.6	+1.21	+1.46
M2	3520	0.032	0.48	0.40	+8.47	−1.89	+10.4	+1.18	+1.49
M3	3400	0.020	0.41	—	+8.97	−2.15	+11.1	+1.16	+1.51
M4	3290	0.013	0.35	—	+9.49	−2.38	+11.9	+1.15	+1.54
M5	3170	0.0076	0.29	0.21	+10.1	−2.73	+12.8	+1.24	+1.64
M6	3030	0.0044	0.24	—	+10.6	−3.21	+13.8	+1.32	+1.73
M7	2860	0.0025	0.20	—	+11.3	−3.46	+14.7	+1.40	+1.80

[a]Values adopted in this text.

Giant Stars (Luminosity Class III)

Sp. Type	T_e (K)	$L/L_\odot$	$R/R_\odot$	$M/M_\odot$	$M_{\rm bol}$	BC	M_V	$U-B$	$B-V$
O5	39400	741000	18.5	—	−9.94	−4.05	−5.9	−1.18	−0.32
O6	37800	519000	16.8	—	−9.55	−3.80	−5.7	−1.17	−0.32
O7	36500	375000	15.4	—	−9.20	−3.58	−5.6	−1.14	−0.32
O8	35000	277000	14.3	—	−8.87	−3.39	−5.5	−1.13	−0.31
B0	29200	84700	11.4	20	−7.58	−2.88	−4.7	−1.08	−0.29
B1	24500	32200	10.0	—	−6.53	−2.43	−4.1	−0.97	−0.26
B2	20200	11100	8.6	—	−5.38	−2.02	−3.4	−0.91	−0.24
B3	18300	6400	8.0	—	−4.78	−1.60	−3.2	−0.74	−0.20
B5	15100	2080	6.7	7	−3.56	−1.30	−2.3	−0.58	−0.17
B6	13800	1200	6.1	—	−2.96	−1.13	−1.8	−0.51	−0.15
B7	12700	710	5.5	—	−2.38	−0.97	−1.4	−0.44	−0.13
B8	11700	425	5.0	—	−1.83	−0.82	−1.0	−0.37	−0.11
B9	10900	263	4.5	—	−1.31	−0.71	−0.6	−0.20	−0.07
A0	10200	169	4.1	4	−0.83	−0.42	−0.4	−0.07	−0.03
A1	9820	129	3.9	—	−0.53	−0.29	−0.2	+0.07	+0.01
A2	9460	100	3.7	—	−0.26	−0.20	−0.1	+0.06	+0.05
A5	8550	52	3.3	—	+0.44	−0.14	+0.6	+0.11	+0.15
A8	7830	33	3.1	—	+0.95	−0.10	+1.0	+0.10	+0.25
F0	7400	27	3.2	—	+1.17	−0.11	+1.3	+0.08	+0.30
F2	7000	24	3.3	—	+1.31	−0.11	+1.4	+0.08	+0.35
F5	6410	22	3.8	—	+1.37	−0.14	+1.5	+0.09	+0.43
G0	5470	29	6.0	1.0	+1.10	−0.20	+1.3	+0.21	+0.65
G2	5300	31	6.7	—	+1.00	−0.27	+1.3	+0.39	+0.77
G8	4800	44	9.6	—	+0.63	−0.42	+1.0	+0.70	+0.94
K0	4660	50	10.9	1.1	+0.48	−0.50	+1.0	+0.84	+1.00
K1	4510	58	12.5	—	+0.32	−0.55	+0.9	+1.01	+1.07
K3	4260	79	16.4	—	−0.01	−0.76	+0.8	+1.39	+1.27
K4	4150	93	18.7	—	−0.18	−0.94	+0.8	—	+1.38
K5	4050	110	21.4	1.2	−0.36	−1.02	+0.7	+1.81	+1.50
K7	3870	154	27.6	—	−0.73	−1.17	+0.4	+1.83	+1.53
M0	3690	256	39.3	1.2	−1.28	−1.25	+0.0	+1.87	+1.56
M1	3600	355	48.6	—	−1.64	−1.44	−0.2	+1.88	+1.58
M2	3540	483	58.5	1.3	−1.97	−1.62	−0.4	+1.89	+1.60
M3	3480	643	69.7	—	−2.28	−1.87	−0.4	+1.88	+1.61
M4	3440	841	82.0	—	−2.57	−2.22	−0.4	+1.73	+1.62
M5	3380	1100	96.7	—	−2.86	−2.48	−0.4	+1.58	+1.63
M6	3330	1470	116	—	−3.18	−2.73	−0.4	+1.16	+1.52

Supergiant Stars (Luminosity Class Approximately Iab)									
Sp. Type	T_e (K)	$L/L_\odot$	$R/R_\odot$	$M/M_\odot$	M_{bol}	BC	M_V	$U-B$	$B-V$
O5	40900	1140000	21.2	70	−10.40	−3.87	−6.5	−1.17	−0.31
O6	38500	998000	22.4	40	−10.26	−3.74	−6.5	−1.16	−0.31
O7	36200	877000	23.8	—	−10.12	−3.48	−6.6	−1.14	−0.31
O8	34000	769000	25.3	28	−9.98	−3.35	−6.6	−1.13	−0.29
B0	26200	429000	31.7	25	−9.34	−2.49	−6.9	−1.06	−0.23
B1	21400	261000	37.3	—	−8.80	−1.87	−6.9	−1.00	−0.19
B2	17600	157000	42.8	—	−8.25	−1.58	−6.7	−0.94	−0.17
B3	16000	123000	45.8	—	−7.99	−1.26	−6.7	−0.83	−0.13
B5	13600	79100	51.1	20	−7.51	−0.95	−6.6	−0.72	−0.10
B6	12600	65200	53.8	—	−7.30	−0.88	−6.4	−0.69	−0.08
B7	11800	54800	56.4	—	−7.11	−0.78	−6.3	−0.64	−0.05
B8	11100	47200	58.9	—	−6.95	−0.66	−6.3	−0.56	−0.03
B9	10500	41600	61.8	—	−6.81	−0.52	−6.3	−0.50	−0.02
A0	9980	37500	64.9	16	−6.70	−0.41	−6.3	−0.38	−0.01
A1	9660	35400	67.3	—	−6.63	−0.32	−6.3	−0.29	+0.02
A2	9380	33700	69.7	—	−6.58	−0.28	−6.3	−0.25	+0.03
A5	8610	30500	78.6	13	−6.47	−0.13	−6.3	−0.07	+0.09
A8	7910	29100	91.1	—	−6.42	−0.03	−6.4	+0.11	+0.14
F0	7460	28800	102	12	−6.41	−0.01	−6.4	+0.15	+0.17
F2	7030	28700	114	—	−6.41	0.00	−6.4	+0.18	+0.23
F5	6370	29100	140	10	−6.42	−0.03	−6.4	+0.27	+0.32
F8	5750	29700	174	—	−6.44	−0.09	−6.4	+0.41	+0.56
G0	5370	30300	202	10	−6.47	−0.15	−6.3	+0.52	+0.76
G2	5190	30800	218	—	−6.48	−0.21	−6.3	+0.63	+0.87
G8	4700	32400	272	—	−6.54	−0.42	−6.1	+1.07	+1.15
K0	4550	33100	293	13	−6.56	−0.50	−6.1	+1.17	+1.24
K1	4430	34000	314	—	−6.59	−0.56	−6.0	+1.28	+1.30
K3	4190	36100	362	—	−6.66	−0.75	−5.9	+1.60	+1.46
K4	4090	37500	386	—	−6.70	−0.90	−5.8	—	+1.53
K5	3990	39200	415	13	−6.74	−1.01	−5.7	+1.80	+1.60
K7	3830	43200	473	—	−6.85	−1.20	−5.6	+1.84	+1.63
M0	3620	51900	579	13	−7.05	−1.29	−5.8	+1.90	+1.67
M1	3490	60300	672	—	−7.21	−1.38	−5.8	+1.90	+1.69
M2	3370	72100	791	19	−7.41	−1.62	−5.8	+1.95	+1.71
M3	3210	89500	967	—	−7.64	−2.13	−5.5	+1.95	+1.69
M4	3060	117000	1220	—	−7.93	−2.75	−5.2	+2.00	+1.76
M5	2880	165000	1640	24	−8.31	−3.47	−4.8	+1.60	+1.80
M6	2710	264000	2340	—	−8.82	−3.90	−4.9	—	—

APPENDIX

H The Messier Catalog

M	NGC	Name	Const.	m_V[a]	R. A.[b] h	R. A.[b] m	Dec.[b] °	Dec.[b] ′	Type[c]
1	1952	Crab	Tau	8.4:	5	34.5	+22	01	SNR
2	7089		Aqr	6.5	21	33.5	−0	49	GC
3	5272		CVn	6.4	13	42.2	+28	23	GC
4	6121		Sco	5.9	16	23.6	−26	32	GC
5	5904		Ser	5.8	15	18.6	+2	05	GC
6	6405		Sco	4.2	17	40.1	−32	13	OC
7	6475		Sco	3.3	17	53.9	−34	49	OC
8	6523	Lagoon	Sgr	5.8:	18	03.8	−24	23	N
9	6333		Oph	7.9:	17	19.2	−18	31	GC
10	6254		Oph	6.6	16	57.1	−4	06	GC
11	6705		Sct	5.8	18	51.1	−6	16	OC
12	6218		Oph	6.6	16	47.2	−1	57	GC
13	6205		Her	5.9	16	41.7	+36	28	GC
14	6402		Oph	7.6	17	37.6	−3	15	GC
15	7078		Peg	6.4	21	30.0	+12	10	GC
16	6611		Ser	6.0	18	18.8	−13	47	OC
17	6618	Swan[d]	Sgr	7:	18	20.8	−16	11	N
18	6613		Sgr	6.9	18	19.9	−17	08	OC
19	6273		Oph	7.2	17	02.6	−26	16	GC
20	6514	Trifid	Sgr	8.5:	18	02.6	−23	02	N
21	6531		Sgr	5.9	18	04.6	−22	30	OC
22	6656		Sgr	5.1	18	36.4	−23	54	GC
23	6494		Sgr	5.5	17	56.8	−19	01	OC
24	6603		Sgr	4.5:	18	16.9	−18	29	OC
25			Sgr	4.6	18	31.6	−19	15	OC
26	6694		Sct	8.0	18	45.2	−9	24	OC
27	6853	Dumbbell	Vul	8.1:	19	59.6	+22	43	PN
28	6626		Sgr	6.9:	18	24.5	−24	52	GC
29	6913		Cyg	6.6	20	23.9	+38	32	OC
30	7099		Cap	7.5	21	40.4	−23	11	GC
31	224	Andromeda	And	3.4	0	42.7	+41	16	SbI–II
32	221		And	8.2	0	42.7	+40	52	cE2
33	598	Triangulum	Tri	5.7	1	33.9	+30	39	Sc(s)II–III
34	1039		Per	5.2	2	42.0	+42	47	OC
35	2168		Gem	5.1	6	08.9	+24	20	OC
36	1960		Aur	6.0	5	36.1	+34	08	OC
37	2099		Aur	5.6	5	52.4	+32	33	OC
38	1912		Aur	6.4	5	28.7	+35	50	OC
39	7092		Cyg	4.6	21	32.2	+48	26	OC
40			UMa	8:	12	22.4	+58	05	DS

M	NGC	Name	Const.	m_V[a]	R. A.[b] h	m	Dec.[b] °	′	Type[c]
41	2287		CMa	4.5	6	47.0	−20	44	OC
42	1976	Orion[e]	Ori	4:	5	35.3	−5	23	N
43	1982		Ori	9:	5	35.6	−5	16	N
44	2632	Praesepe	Cnc	3.1	8	40.1	+19	59	OC
45		Pleiades	Tau	1.2	3	47.0	+24	07	OC
46	2437		Pup	6.1	7	41.8	−14	49	OC
47	2422		Pup	4.4	7	36.6	−14	30	OC
48	2548		Hya	5.8	8	13.8	−5	48	OC
49	4472		Vir	8.4	12	29.8	+8	00	E2
50	2323		Mon	5.9	7	03.2	−8	20	OC
51	5194	Whirlpool[f]	CVn	8.1	13	29.9	+47	12	Sbc(s)I–II
52	7654		Cas	6.9	23	24.2	+61	35	OC
53	5024		Com	7.7	13	12.9	+18	10	GC
54	6715		Sgr	7.7	18	55.1	−30	29	GC
55	6809		Sgr	7.0	19	40.0	−30	58	GC
56	6779		Lyr	8.2	19	16.6	+30	11	GC
57	6720	Ring	Lyr	9.0:	18	53.6	+33	02	PN
58	4579		Vir	9.8	12	37.7	+11	49	Sab(s)II
59	4621		Vir	9.8	12	42.0	+11	39	E5
60	4649		Vir	8.8	12	43.7	+11	33	E2
61	4303		Vir	9.7	12	21.9	+4	28	Sc(s)I
62	6266		Oph	6.6	17	01.2	−30	07	GC
63	5055	Sunflower	CVn	8.6	13	15.8	+42	02	Sbc(s)II–III
64	4826	Evil Eye	Com	8.5	12	56.7	+21	41	Sab(s)II
65	3623		Leo	9.3	11	18.9	+13	05	Sa(s)I
66	3627		Leo	9.0	11	20.2	+12	59	Sb(s)II
67	2682		Cnc	6.9	8	50.4	+11	49	OC
68	4590		Hya	8.2	12	39.5	−26	45	GC
69	6637		Sgr	7.7	18	31.4	−32	21	GC
70	6681		Sgr	8.1	18	43.2	−32	18	GC
71	6838		Sge	8.3	19	53.8	+18	47	GC
72	6981		Aqr	9.4	20	53.5	−12	32	GC
73	6994		Aqr	9.1	20	58.9	−12	38	OC
74	628		Psc	9.2	1	36.7	+15	47	Sc(s)I
75	6864		Sgr	8.6	20	06.1	−21	55	GC
76	650/651		Per	11.5:	1	42.3	+51	34	PN
77	1068		Cet	8.8	2	42.7	−0	01	Sb(rs)II
78	2068		Ori	8:	5	46.7	+0	03	N
79	1904		Lep	8.0	5	24.5	−24	33	GC
80	6093		Sco	7.2	16	17.0	−22	59	GC
81	3031		UMa	6.8	9	55.6	+69	04	Sb(r)I–II
82	3034		UMa	8.4	9	55.8	+69	41	Amorph
83	5236		Hya	7.6:	13	37.0	−29	52	SBc(s)II
84	4374		Vir	9.3	12	25.1	+12	53	E1
85	4382		Com	9.2	12	25.4	+18	11	S0 pec
86	4406		Vir	9.2	12	26.2	+12	57	S0/E3
87	4486	Virgo A	Vir	8.6	12	30.8	+12	24	E0
88	4501		Com	9.5	12	32.0	+14	25	Sbc(s)II
89	4552		Vir	9.8	12	35.7	+12	33	S0
90	4569		Vir	9.5	12	36.8	+13	10	Sab(s)I–II

M	NGC	Name	Const.	m_V[a]	R. A.[b] h	m	Dec.[b] °	′	Type[c]
91	4548		Com	10.2	12	35.4	+14	30	SBb(rs)I–II
92	6341		Her	6.5	17	17.1	+43	08	GC
93	2447		Pup	6.2:	7	44.6	−23	52	OC
94	4736		CVn	8.1	12	50.9	+41	07	RSab(s)
95	3351		Leo	9.7	10	44.0	+11	42	SBb(r)II
96	3368		Leo	9.2	10	46.8	+11	49	Sab(s)II
97	3587	Owl	UMa	11.2:	11	14.8	+55	01	PN
98	4192		Com	10.1	12	13.8	+14	54	SbII
99	4254		Com	9.8	12	18.8	+14	25	Sc(s)I
100	4321		Com	9.4	12	22.9	+15	49	Sc(s)I
101	5457	Pinwheel	UMa	7.7	14	03.2	+54	21	Sc(s)I
102	5866		UMa	10.5	15	06.5	+55	46	S0
103	581		Cas	7.4:	1	33.2	+60	42	OC
104	4594	Sombrero	Vir	8.3	12	40.0	−11	37	Sa/Sb
105	3379		Leo	9.3	10	47.8	+12	35	E0
106	4258		CVn	8.3	12	19.0	+47	18	Sb(s)II
107	6171		Oph	8.1	16	32.5	−13	03	GC
108	3556		UMa	10.0	11	11.5	+55	40	Sc(s)III
109	3992		UMa	9.8	11	57.6	+53	23	SBb(rs)I
110	205		And	8.0	0	40.4	+41	41	S0/E pec

[a] : indicates approximate apparent visual magnitude.

[b] Right ascension and declination are given in epoch 2000.0.

[c] Type abbreviations correspond to: SNR = supernova remnant, GC = globular cluster, OC = open cluster, N = diffuse nebula, PN = planetary nebula, DS = double star. Galaxies are indicated by their morphological Hubble types.

[d] M17, the Swan nebula, is also known as the Omega nebula.

[e] M42 also corresponds to the Trapezium H II region.

[f] M51 also includes NGC 5195, the satellite to the Whirlpool galaxy.

APPENDIX

I Constants, A Programming Module

Constants is a Fortran 95 module implementation of the astronomical and physical constants data given in Appendix A (a C++ header file version is also available). Constants also includes high-precision values of mathematical constants (π, e) and conversion factors between degrees and radians. In addition, Constants provides various machine constants characteristic of the particular platform that a code is running on. For example, in the Fortran 95 implementation, Constants includes machine-queried KIND designations for single, double, and quadruple precision, the smallest and largest numbers that can be represented by the computer for a specific precision, and the number of significant figures that can be represented for each level of precision.

Constants is employed by the other codes described in the appendices that follow.

The source code is available for download from the resources tab on following webpage: www.cambridge.org/astrophysics.

APPENDIX J

`Orbit`, A Planetary Orbit Code

`Orbit` is a computer program designed to calculate the position of a planet orbiting a massive star (or, alternatively, the orbit of the reduced mass about the center of mass of the system). The program is based on Kepler's laws of planetary motion as derived in Chapter 2. References to the relevant equations are given in the comment sections of the code.

The user is asked to enter the mass of the parent star (in solar masses), the semimajor axis of the orbit (in AU), and the eccentricity of the orbit. The user is also asked to enter the number of time steps desired for the calculation (perhaps 1000 to 100,000) and the frequency with which the time steps are to be printed to the output file (`Orbit.txt`). If 1000 time steps are specified with a frequency of 10, then 100 evenly spaced (in time) time steps will be printed.

The output file can be imported directly into a graphics or spreadsheet program in order to generate a graph of the orbit. Note that it may be necessary to delete the header information in `Orbit.txt` prior to importing the data columns into the graphics or spreadsheet program.

The source code is available in both Fortran 95 and C++ versions. Compiled versions of the code are also available.

The code may be downloaded from the companion website at www.cambridge.org/astrophyisics.

APPENDIX K

TwoStars, A Binary Star Code

As discussed in Chapter 7, binary star systems play a very important role in determining various stellar properties, including masses and radii. In addition, analyses using sophisticated binary-star modeling codes can provide information about variations in surface flux such as limb darkening (see Section 9.3) and the presence of star spots (e.g., Section 11.3) or reflective heating. Advanced codes can also detail the effects of gravitational tidal interactions and centrifugal forces that result in stars that deviate (sometimes significantly) from spherical symmetry; refer to Chapter 18.

A simple binary star code is developed in this appendix that incorporates a number of the basic features of more sophisticated codes. `TwoStars` is designed to provide position, radial-velocity, and binary light curve information that can be used to determine masses (m_1 and m_2 from determination of the semimajor axes and periods of the orbits), radii (R_1 and R_2 by measuring eclipse times), effective temperature ratios (from the relative depths of the primary and secondary minima), limb darkening, orbital eccentricity (e), orbital inclination (i), and orientation of periastron (ϕ). However, in order to greatly simplify the code, it is assumed that the two stars are strictly spherically symmetric, that they do not collide with one another, and that their surface fluxes vary only with stellar radius (i.e., there are no anomalous star spots or localized heating).

To begin, assume that the orbits of the two stars lie in the x–y plane with the center of mass of the system located at the origin of the coordinate system, as shown in Fig. K.1 (the z-axis is out of the page). In order to generalize the orientation of the orbit, periastron for Star 1 (the point in the orbit closest to the center of mass) is at an angle ϕ measured counterclockwise from the positive x-axis and in the direction of the orbital motion. It is also assumed that the orbital plane is inclined an angle i with respect to the plane of the sky (the y'–z' plane) as shown in Fig. K.2. The line of sight from the observer to the center of mass is along the x'-axis, and the center of mass is located at the origin of the primed coordinate system. Finally, the y'-axis is directed out of the page and is aligned with the y-axis of Fig. K.1.

It is a straightforward process to show that the transformation between the two coordinate systems is given by

$$x' = z\cos i + x\sin i \tag{K.1}$$

$$y' = y \tag{K.2}$$

$$z' = z\sin i - x\cos i, \tag{K.3}$$

which of course simplifies significantly for the case where the centers of mass lie along the x–y plane (i.e., $z = 0$).

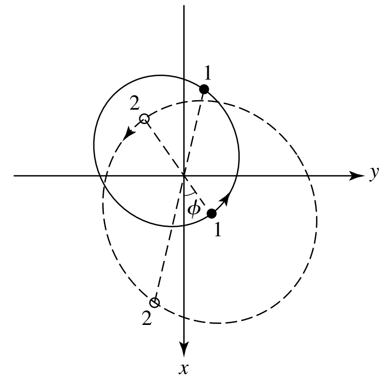

FIGURE K.1 The orbits of the stars in the binary system lie in the x–y plane, with the z-axis directed out of the page. The center of mass of the system is located at the origin of the coordinate system. In this example $m_2/m_1 = 0.68$, $e = 0.4$, and $\phi = 35°$. The two positions of Stars 1 and 2 are separated by $P/4$, where P is the orbital period.

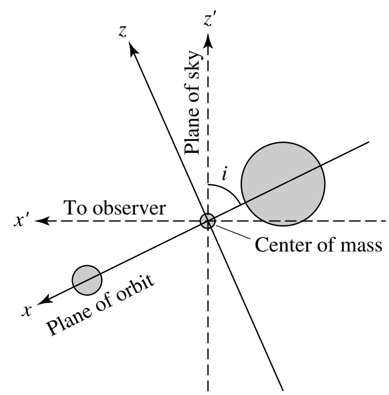

FIGURE K.2 The plane of the orbit is inclined an angle i with respect to the plane of the sky (the y'–z' plane). The line of sight from the observer to the center of mass is along the x'-axis, and the center of mass is located at the origin of the primed coordinate system. The y'-axis is aligned with the y-axis of Fig. K.1, and both are directed out of the page. The foreground star in this illustration is the smaller star.

The motions of the stars in the x–y plane are determined directly by using Kepler's laws and invoking the concept of the reduced mass, as described in Chapter 2. The approach is similar to what was used in `Orbit`, Appendix J, except that no assumption is made about the relative masses of the two objects in the system [in `Orbit` it was assumed that one object (a planet) was much less massive than the other object (the parent star)].

A careful reading of the code available on the companion website will identify several explicit instances of plus and minus signs associated with the variables `vr`, `v1r`, `v2r`, `x1`, `y1`, `x2`, and `y2`. The choice of minus signs corresponds to the choice of the coordinate system and its relationship to the observer. For instance, if the inclination angle is $i = 90°$, then

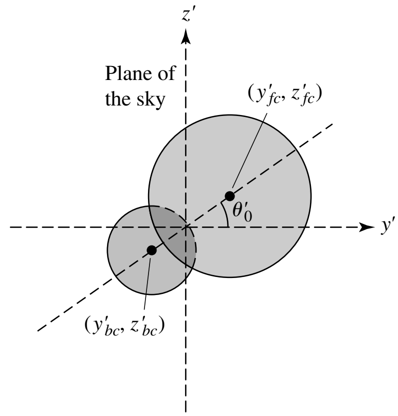

FIGURE K.3 The disks of the two stars projected onto the plane of the sky. The foreground star (assumed in this illustration to be the larger star) has $x'_c > 0$, where x'_c is the x' coordinate of its center of mass. The angle between the y'-axis and the line connecting the centers of the disks of the two stars projected onto the plane of the sky is θ'_0.

the x and x' axes are aligned and motion in the positive x direction corresponds to motion toward the observer (a negative radial velocity).

In order to compute the light curve for the eclipse, it is necessary to integrate the luminous flux over the portion of each star's surface that is visible to the observer. This is done by first determining which star is in front of the other. Given that the plane of the sky corresponds to the y'–z' plane, the center of mass of the star that is closest to the observer has the coordinate value $x' > 0$ (see Fig. K.3).

If the star in front is partially or entirely eclipsing the background star, then the distances between their centers of mass projected onto the y'–z' plane must be less than the sum of their radii; or, for an eclipse to be taking place,

$$\sqrt{\left(y'_{fc} - y'_{bc}\right)^2 + \left(z'_{fc} - z'_{bc}\right)^2} < R_f + R_b, \tag{K.4}$$

where (y'_{fc}, z'_{fc}) and (y'_{bc}, z'_{bc}) are the locations of the centers of mass of the foreground and background stars, respectively, as projected onto the plane of the sky.

To optimize the computation of the integrated luminous flux, it is appropriate to locate the line of symmetry between the centers of mass of the two stars. Again referring to Fig. K.3, we see that the angle between the y'-axis and the line connecting the projected centers of mass is given by

$$\theta'_0 = \tan^{-1}\left(\frac{z'_{fc} - z'_{bc}}{y'_{fc} - y'_{bc}}\right). \tag{K.5}$$

Once the background star has been identified and the line of symmetry determined, the decrease in the amount of light due to the eclipse can be computed by first finding out which parts of the background star are behind the foreground star. If a point on the eclipsed disk is within a distance R_f of the center of the foreground star's disk as projected onto the y'–z'

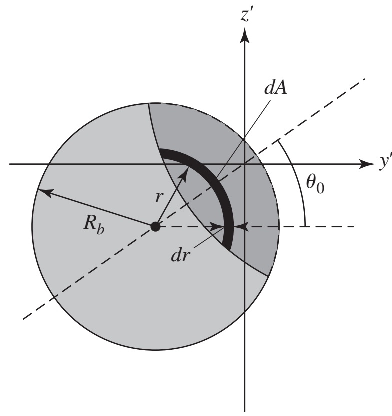

FIGURE K.4 The region of the background star being eclipsed is shown in dark gray. Numerical integration of the flux over arcs of various radii r and thickness dr makes it possible to determine how much light is blocked by the foreground star.

plane, then that point on the star's surface is behind the foreground star. In other words, the condition for a point (y'_b, z'_b) on the disk of the background star to be behind the disk of the foreground star is

$$\sqrt{\left(y'_b - y'_{fc}\right)^2 + \left(z'_b - z'_{fc}\right)^2} < R_f. \tag{K.6}$$

The eclipsed region can then be mapped out by starting along the line of symmetry at some distance r from the center of the disk of the background star and moving at increasing angles of $\Delta\theta'$ from θ'_0 until the inequality of Eq. (K.6) is no longer satisfied or until $\Delta\theta'$ exceeds 180°. In the later case, this would imply that the entire disk within the radius r of its center is eclipsed. Given the assumption of spherical symmetry, the region of the background star's disk between $\theta'_{\rm min} = -\Delta\theta' + \theta'_0$ and θ'_0 is identical to the region between θ'_0 and $\theta'_{\rm max} = \Delta\theta' + \theta'_0$ for a fixed value of r (see Fig. K.4). For an arc-shaped surface of radius r and width dr, the area of the surface is given by

$$dA = r\,dr\,\left(\theta'_{\rm max} - \theta'_{\rm min}\right) = 2r\,dr\,\Delta\theta'. \tag{K.7}$$

Now, if the luminous flux at that radius from the center of the background star's disk is $F(r)$, the amount of light in that arc that has been blocked is given by

$$dS = F(r)\,dA = 2F(r)\,r\,dr\,\Delta\theta'. \tag{K.8}$$

By subtracting the loss in light due to each eclipsed arc from the total light of the uneclipsed star, we can determine the total amount of light received from the background star during a partial or total eclipse. (Note that due to the effects of limb darkening, $F(r)$ is not constant across the entire disk.)

Finally, all that remains is to convert the total amount of light received to magnitudes (see Eq. 3.8).

TwoStars implements each of the ideas described. An example of the input required for TwoStars, along with the first ten lines of model output, is shown in Fig. K.5.

The source code for (TwoStars), together with compiled versions of the program, is available for download from the resources tab on the following webpage: www.cambridge.org/astrophysics.

```
Specify the name of your output file: c:\YYSgr.txt


Enter the data for Star #1
   Mass (solar masses):             5.9
   Radius (solar radii):            3.2
   Effective Temperature (K):       15200

Enter the data for Star #2
   Mass (solar masses):             5.6
   Radius (solar radii):            2.9
   Effective Temperature (K):       13700

Enter the desired orbital parameters
   Orbital Period (days):           2.6284734
   Orbital Eccentricity:            0.1573
   Orbital Inclination (deg):       88.89
   Orientation of Periastron (deg): 214.6

Enter the x', y', and z' components of the center of mass velocity vector:
Notes:  (1)  The plane of the sky is (y',z')
        (2)  If v_x' < 0, then the center of mass is blueshifted

   v_x' (km/s)                      0
   v_y' (km/s)                      0
   v_z' (km/s)                      0


The semimajor axis of the reduced mass is     0.084318 AU
   a1 =      0.040971 AU
   a2 =      0.043166 AU

         t/P          v1r (km/s)       v2r (km/s)          Mbol        dS (W)
       0.000000       112.824494      -118.868663       -2.457487   0.000000E+00
       0.000999       114.247263      -120.367652       -2.457487   0.000000E+00
       0.001998       115.660265      -121.856351       -2.457487   0.000000E+00
       0.002997       117.063327      -123.334576       -2.457487   0.000000E+00
       0.003996       118.456275      -124.802147       -2.457487   0.000000E+00
       0.004995       119.838940      -126.258884       -2.457487   0.000000E+00
       0.005994       121.211155      -127.704610       -2.457487   0.000000E+00
       0.006993       122.572755      -129.139152       -2.457487   0.000000E+00
       0.007992       123.923575      -130.562338       -2.457487   0.000000E+00
       0.008991       125.263455      -131.973998       -2.457487   0.000000E+00
```

FIGURE K.5 An example of the input required for the Fortran 95 command-line version of TwoStars for the system YY Sgr (Fig. 7.2). The first ten lines of model output to the screen are also shown.

APPENDIX L

StatStar, A Stellar Structure Code

StatStar is based on the equations of stellar structure and the constitutive relations developed in Chapters 9 and 10. An example of the output generated by StatStar is available on the companion website.

StatStar is designed to illustrate as clearly as possible many of the most important aspects of numerical stellar astrophysics. To accomplish this goal, StatStar models are restricted to a fixed composition throughout [in other words, they are homogeneous zero-age main-sequence models (ZAMS)].

The four basic stellar structure equations are computed in the functions dPdr (Eq. 10.6), dMdr (Eq. 10.7), dLdr (Eq. 10.36), and dTdr [either Eq. (10.68) or Eq. (10.89), for radiation or adiabatic convection, respectively].

The density [$\rho(r) =$ rho] is calculated directly from the ideal gas law and the radiation pressure equation (Eq. 10.20) in FUNCTION Opacity, given local values for the pressure [$P(r) =$ P], temperature [$T(r) =$ T], and mean molecular weight ($\mu =$ mu, assumed here to be for a completely ionized gas only). Once the density is determined, both the opacity [$\overline{\kappa}(r) =$ kappa] and the nuclear energy generation rate [$\epsilon(r) =$ epsilon] are calculated. The opacity is determined in FUNCTION Opacity using the bound–bound and bound–free opacity formulae [Eqs. (9.22) and (9.23), respectively], together with electron scattering (Eq. 9.27) and H^- ion (Eq. 9.28) contributions. The energy generation rate is calculated in Function Nuclear from the equations for the total pp chain (Eq. 10.46) and the CNO cycle (Eq. 10.58).[1]

The program begins by asking the user to supply the desired stellar mass (Msolar, in solar units), the trial effective temperature (Teff, in kelvins), the trial luminosity (Lsolar, also in solar units), and the mass fractions of hydrogen (X) and metals (Z). Using the stellar structure equations, the program proceeds to integrate from the surface of the star toward the center, stopping when a problem is detected or when a satisfactory solution is obtained. If the inward integration is not successful, a new trial luminosity and/or effective temperature must be chosen. Recall from page 333 that the Vogt–Russell theorem states that a unique stellar structure exists for a given mass and composition. Satisfying the central boundary conditions therefore requires specific surface boundary conditions. It is for this reason that a well-defined main sequence exists.

Since it is nearly impossible to satisfy the central boundary conditions exactly by the crude *shooting method* employed by StatStar, the calculation is terminated when the core is approached. The stopping criteria used here are that the interior mass $M_r < M_{\rm min}$ and the interior luminosity $L_r < L_{\rm min}$, when the radius $r < R_{\rm min}$, where $M_{\rm min}$, $L_{\rm min}$, and

[1] State-of-the-art research codes use much more sophisticated prescriptions for the equations of state.

$R_{\rm min}$ are specified as fractions of the surface mass (M_s), luminosity (L_s), and radius (R_s), respectively. Once the criteria for halting the integration are detected, the conditions at the center of the star are estimated by an extrapolation procedure.

`StatStar` makes the simplifying assumptions that the pressure, temperature, and density are all zero at the surface of the star. As a result, it is necessary to begin the calculation with approximations to the basic stellar structure equations. This can be seen by noting that the mass, pressure, luminosity, and temperature gradients are all proportional to the density and are therefore exactly zero at the surface. Curiously, it would appear that applying these gradients in their usual form implies that the fundamental physical parameters cannot change from their initial values, since the density would remain zero at each step!

One way to overcome this problem is to assume that the interior mass and luminosity are both constant through a small number of surface zones. In the case of the luminosity, this is clearly a valid assumption since temperatures are not sufficient to produce nuclear reactions near the surfaces of ZAMS stars; and furthermore, since ZAMS stars are static, changes in gravitational potential energy are necessarily zero. For the interior mass, the assumption is not quite as obvious. However, we will see that in realistic stellar models, the density is so low near the surface that the approximation is indeed very reasonable. Of course, it is important to verify that the assumptions are not being violated to within specified limits.

Given the surface values $M_r = M_s$ and $L_r = L_s$, and assuming that the surface zone is radiative, dividing Eq. (10.6) by Eq. (10.68) leads to

$$\frac{dP}{dT} = \frac{16\pi ac}{3}\frac{GM_s}{L_s}\frac{T^3}{\overline{\kappa}}.$$

Since relatively few free electrons exist in the thin outer atmospheres of stars, electron scattering and H^- ion contributions to the opacity will be neglected in the surface zone approximation. In this case $\overline{\kappa}$ may be replaced by the bound–free and free–free Kramers opacity laws [Eqs. (9.22) and (9.23)], expressed in the forms $\overline{\kappa}_{bf} = A_{bf}\rho/T^{3.5}$ and $\overline{\kappa}_{ff} = A_{ff}\rho/T^{3.5}$, respectively. Defining $A \equiv A_{bf} + A_{ff}$ and using Eq. (10.11) to express the density in terms of the pressure and temperature through the ideal gas law (assuming that radiation pressure may be neglected), we get

$$\frac{dP}{dT} = \frac{16\pi}{3}\frac{GM_s}{L_s}\frac{ack}{A\mu m_H}\frac{T^{7.5}}{P}.$$

Integrating with respect to temperature and solving for the pressure, we find that

$$P = \left(\frac{1}{4.25}\frac{16\pi}{3}\frac{GM_s}{L_s}\frac{ack}{A\mu m_H}\right)^{1/2} T^{4.25}. \tag{L.1}$$

It is now possible to write T in terms of the independent variable r through Eq. (10.68), again using the ideal gas law and Kramers opacity laws, along with Eq. (L.1) to eliminate the dependence on pressure. Integrating yields

$$T = GM_s\left(\frac{\mu m_H}{4.25k}\right)\left(\frac{1}{r} - \frac{1}{R_s}\right). \tag{L.2}$$

Equation (L.2) is first used to obtain a value for $T(r)$; then Eq. (L.1) gives $P(r)$. At this point it is possible to calculate ρ, $\overline{\kappa}$, and ϵ from the usual equation-of-state routines.

A very similar procedure is used in the case that the surface is convective. In this situation, Eq. (10.89) may be integrated directly if γ is constant. This gives

$$T = GM_s\left(\frac{\gamma - 1}{\gamma}\right)\left(\frac{\mu m_H}{k}\right)\left(\frac{1}{r} - \frac{1}{R_s}\right). \tag{L.3}$$

Now, since convection is assumed to be adiabatic in the interior of our simple model, the pressure may be found from Eq. (10.83). The routine `Surface` computes Eqs. (L.1), (L.2), and (L.3).

The conditions at the center of the star are estimated by extrapolating from the last zone that was calculated by direct numerical integration. Beginning with Eq. (10.6) and identifying $M_r = 4\pi\rho_0 r^3/3$, where ρ_0 is taken to be the average density of the central ball (the region inside the last zone calculated by the usual procedure),[2] we get

$$\frac{dP}{dr} = -G\frac{M_r\rho_0}{r^2} = -\frac{4\pi}{3}G\rho_0^2\, r.$$

Integrating yields

$$\int_{P_0}^{P} dP = -\frac{4\pi}{3}G\rho_0^2 \int_0^r r\, dr,$$

and solving for the central pressure results in

$$P_0 = P + \frac{2\pi}{3}G\rho_0^2 r^2.$$

Other central quantities can now be found more directly. Specifically, the central density is estimated to be $\rho_0 = M_r/(4\pi r^3/3)$, where M_r and r are the values of the last zone calculated. T_0 is determined from the ideal gas law and radiation pressure using an iterative procedure (the Newton–Raphson method). Finally, the central value for the nuclear energy generation rate is computed using $\epsilon_0 = L_r/M_r$.

The numerical integration technique employed here is a Runge–Kutta algorithm. The Runge–Kutta algorithm evaluates derivatives at several intermediate points between mass shell boundaries to significantly increase the accuracy of the numerical integration. Details of the algorithm will not be discussed further here; you are encouraged to consult Press, Teukolsky, Vetterling, and Flannery (1996), for details of the implementation.

The source code, together with compiled versions of the program, is available for download from the resources tab on following webpage www.cambridge.org/astrophysics.

[2]You might notice that dP/dr goes to zero as the center is approached. This behavior is indicative of the smooth nature of the solution. Close inspection of the graphs in Section 11.1 showing the detailed interior structure of the Sun illustrates that the first derivatives of many physical quantities go to zero at the center.

APPENDIX

M `Galaxy`, A Tidal Interaction Code

`Galaxy` is a program that calculates the gravitational effect of the close passage of two galactic nuclei on a disk of stars. It is adapted from a program written by M. C. Schroeder and Neil F. Comins and published in *Astronomy* magazine. This program is very similar to the one used by Alar and Juri Toomre in 1972 to perform their ground-breaking studies of the effect of violent tides between galaxies.

In the program there are two galactic nuclei of masses M_1 and M_2. They are treated as point masses, and they move under the influence of their mutual gravitational attraction. To speed the calculations, only M_1 is surrounded by a disk of stars, with the stars initially in circular Keplerian orbits. The gravitational influence of the stars is neglected, meaning that they do not affect the motions of the nuclei or one another. There is no dynamical friction, and so the nuclei follow the simple two-body trajectories that were discussed in Chapter 2. One advantage of a non-self-gravitating disk is that results do not depend on the number of stars in the disk. You can experiment, changing the initial conditions by using just a few stars for a faster running time and then increasing the number of stars to see more detail. The stars respond only to the gravitational pull of the two nuclei.

The goal is to calculate the positions of the nuclei and stars through a number of time steps separated by a time interval Δt. Let the positions of the nuclei at time step i be

$$[X_1(i), Y_1(i), Z_1(i)] \quad \text{and} \quad [X_2(i), Y_2(i), Z_2(i)],$$

and let the position of a star be[1]

$$[x(i), y(i), z(i)].$$

Also, let the velocities of the nuclei and the star be

$$[V_{1,x}(i-1/2), V_{1,y}(i-1/2), V_{1,z}(i-1/2)],$$
$$[V_{2,x}(i-1/2), V_{2,y}(i-1/2), V_{2,z}(i-1/2)],$$

and

$$[v_x(i-1/2), v_y(i-1/2), v_z(i-1/2)].$$

[1]The results do not change with the number of stars used, so one star is enough to illustrate the procedure.

The velocities are the average velocities between the present (i) and the previous ($i - 1$) time steps, so

$$v_x\left(i - \frac{1}{2}\right) = \frac{x(i) - x(i-1)}{\Delta t} \tag{M.1}$$

is the x component of the star's velocity. In a similar manner, the x component of the star's acceleration is

$$a_x(i) = \frac{v_x(i + 1/2) - v_x(i - 1/2)}{\Delta t}. \tag{M.2}$$

Thus the positions and accelerations, which are determined at each time step, "leapfrog" over the velocities, which are determined between time steps.

Given these values of the positions and velocities, the program calculates the x components of the positions and velocities for the next time step in the following way:

1. Find the star's acceleration at the present time step i, using Newton's law of gravity, Eq. (2.11):

$$a_x(i) = \frac{GM_1}{r_1^3(i)}\left[X_1(i) - x(i)\right] + \frac{GM_2}{r_2^3(i)}\left[X_2(i) - x(i)\right], \tag{M.3}$$

where $r_1(i)$ is the distance between the star and M_1 at time step i,

$$r_1(i) = \sqrt{[X_1(i) - x(i)]^2 + [Y_1(i) - y(i)]^2 + [Z_1(i) - z(i)]^2 + s_f^2}, \tag{M.4}$$

and similarly for $r_2(i)$.

Note that because the nuclei and stars are treated as points, their separations could become very small, even zero (although the conservation of angular momentum makes this rather unlikely). As a result, arbitrarily large values of $1/r_1^3$ and $1/r_2^3$ could cause a numerical overflow and bring a lengthy calculation to an abrupt halt. To avoid this numerical disaster, a *softening factor*, s_f, has been included in the calculations of all separations. This is the smallest separation permitted by the program. Its value is large enough to prevent an overflow, but small enough to have little effect on the overall results.

2. Find the star's average velocity at $i + 1/2$,

$$v_x\left(i + \frac{1}{2}\right) = v_x\left(i - \frac{1}{2}\right) + a_x(i)\Delta t. \tag{M.5}$$

3. Find the star's position at the next time step $i + 1$, using

$$x(i+1) = x(i) + v_x\left(i + \frac{1}{2}\right)\Delta t. \tag{M.6}$$